

sTMS Interface Functional Specification

Institute of Technical Education – OMR
Scanning System

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/Updated By
0.1	12/07/2022	Initial version of ITE OMR Scanning System IFS	AvePoint
0.2	20/07/2022	Revised comments based on ITE's feedback	AvePoint
0.3	17/08/2022	Revised interface specification based on ITE's feedback	AvePoint
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Document Acceptance Form

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1. Introduction

1.1 Purpose of this Document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Institute of Technical Education (ITE) OMR Scanning system for students' scanned result and module information.

1.2 Intended Audience

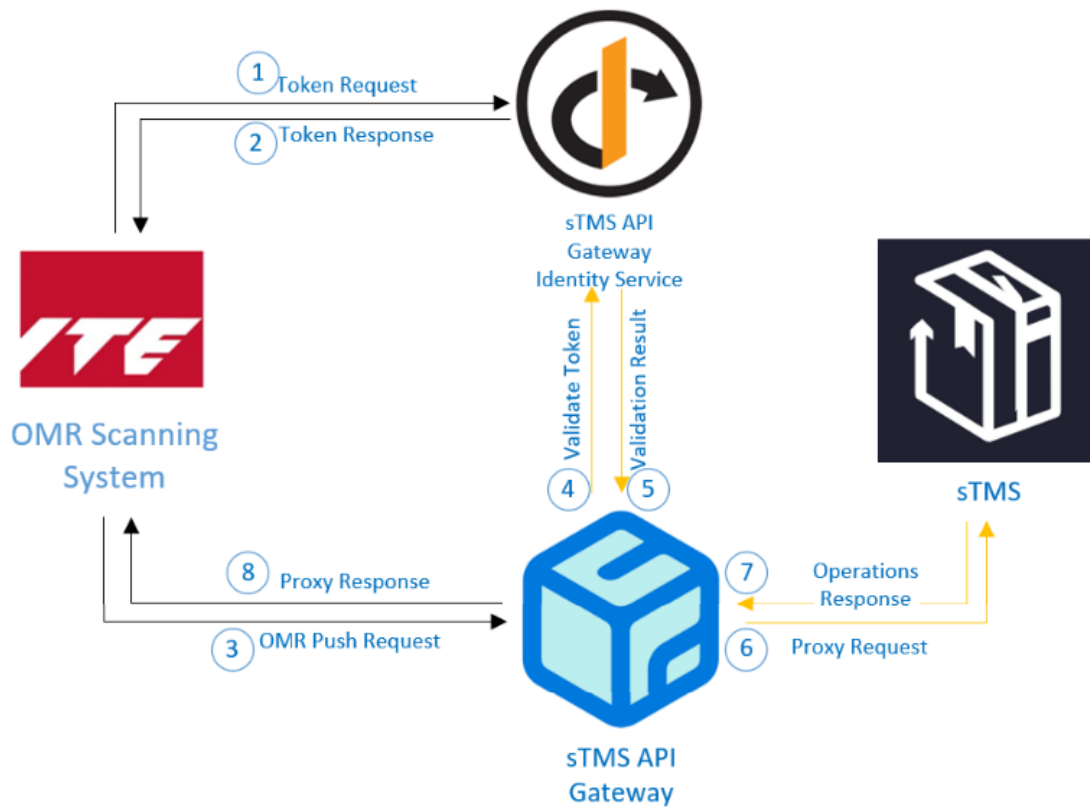
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders, business users and system owners of both systems.

2. Technical Overview

2.1 System Flow

There are 4 roles involved in the interface diagram.

- **ITE OMR Scanning System:** Institute of Technical Education OMR Scanning system.
- **sTMS API Gateway:** sTMS interface Integration middleware that validates HTTP request authentication, authorization, and payload inspection.
- **sTMS API Gateway Identity Service:** sTMS interface Identity Issuer that issues access token for API authentication and authorization.
- **sTMS:** shared Training Management System.



SN	Description
1	ITE OMR Scanning System sends POST request with <i>client_id</i> and <i>secret</i> to sTMS API Gateway Identity Service to get API access token.
2	sTMS API Gateway Identity Service sends the access token back to ITE OMR Scanning System within response message. This access token can be reused within 1 hour, client application can use an access token for multiple API request before the access token expires, after that client application needs to request a new access token again.
3	ITE OMR Scanning System sends HTTP POST request with OMR Result pushed results and reference date to sTMS API Gateway.
4	sTMS API Gateway sends request to sTMS API Gateway Identity Service to validate the access token of the operation response.
5	sTMS API Gateway Identity Service sends back the validation result.
6	sTMS API Gateway verifies the request and sends a proxy request to sTMS.
7	sTMS insert OMR Result into database and send operation result.
8	sTMS API Gateway sends the operation response to OMR Scanning System in the attached file in. 'Successful' field of every record indicates if record score is uploaded successfully or not in the response body.

2.2 Interface Design Summary

Interface Name	SendOMRresults
Hosting System	shared Training Management System
Data Source System	ITE OMR Scanning System
Data Destination System	shared Training Management System
Mode of Interface	HTTP API
Method	POST
Content-Type	application/json
Authorization	Bearer <access_token>
Frequency	Realtime

Encryption Method	HTTPS
Endpoint URL	<TBC>

2.3 Interface Authentication

sTMS use OAuth (Oauth2.0) to perform authentication. ITE OMR Scanning System needs to send HTTP POST request to sTMS Identity Service endpoint to get access token and put the access token into HTTP request 'Authorization' to consume target API.

sTMS Identity Service Endpoints:

Env	URL
UAT	<TBC>
Production	<TBC>

To access the API, the client application needs to authenticate using the client credential grant from API gateway for an access token with the following parameters:

- **grant_type**: REQUIRED. The value must be set to "client_credentials"
- **client_id**: REQUIRED. Client ID.
- **client_secret**: REQUIRED. Client Secret.

Sample:

```
curl -location -request POST '{URL}' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--data-urlencode 'grant_type=client_credentials' \
--data-urlencode 'client_id={client_id}' \
--data-urlencode 'client_secret={client_secret}'
```

Response Body:

```
{
  "access_token": "{access token}",
  "expires_in": 3600,
  "token_type": "Bearer",
```



```
"scope": "{api scope}"  
}
```

2.4 Business Rule

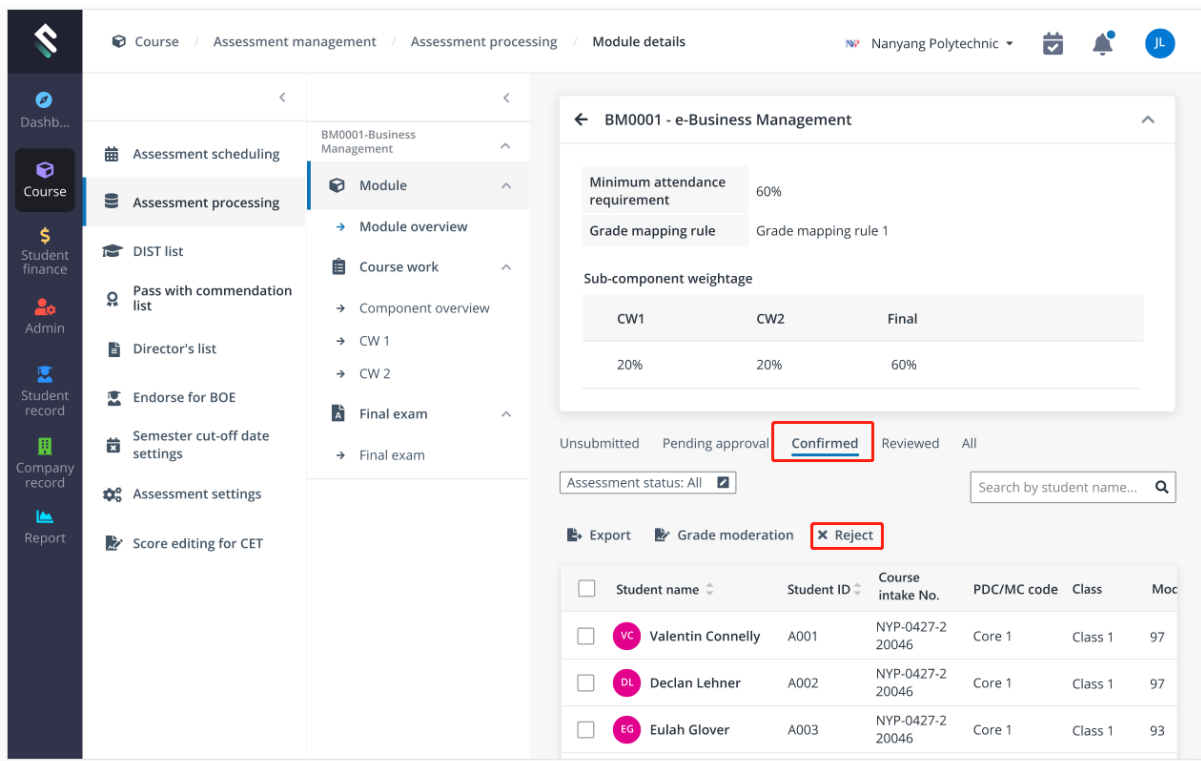
2.4.1 Modules identification

Course Type	Business Rule
Semester-based course	For semester based course, it shall be identified by data fields, which include Student ID, Exam Series, Module Code and Subcomponent
Non-Semester based course	It shall be identified by Student ID, Module Code and Subcomponent

*Please note that **Module Code** must be unique for identification purposes.

2.4.2 Processing results on sTMS

1. The OMR Scanning results sent over shall be recorded as "Confirmed" status in sTMS, the results uploaded to sTMS and amending of results would not be allowed.
For any changes to the results, CET user would have to reject subcomponent score/grade under Module Overview to change the status from 'Confirmed' to 'Unsubmitted'. Only with scores in the 'Unsubmitted' status, the original scores sent from DOSS can be replaced by the subsequent uploads by user.



Course / Assessment management / Assessment processing / Module details

Nanyang Polytechnic

BM0001 - e-Business Management

Minimum attendance requirement: 60%

Grade mapping rule: Grade mapping rule 1

Sub-component weightage

CW1	CW2	Final
20%	20%	60%

Unsubmitted Pending approval **Confirmed** Reviewed All

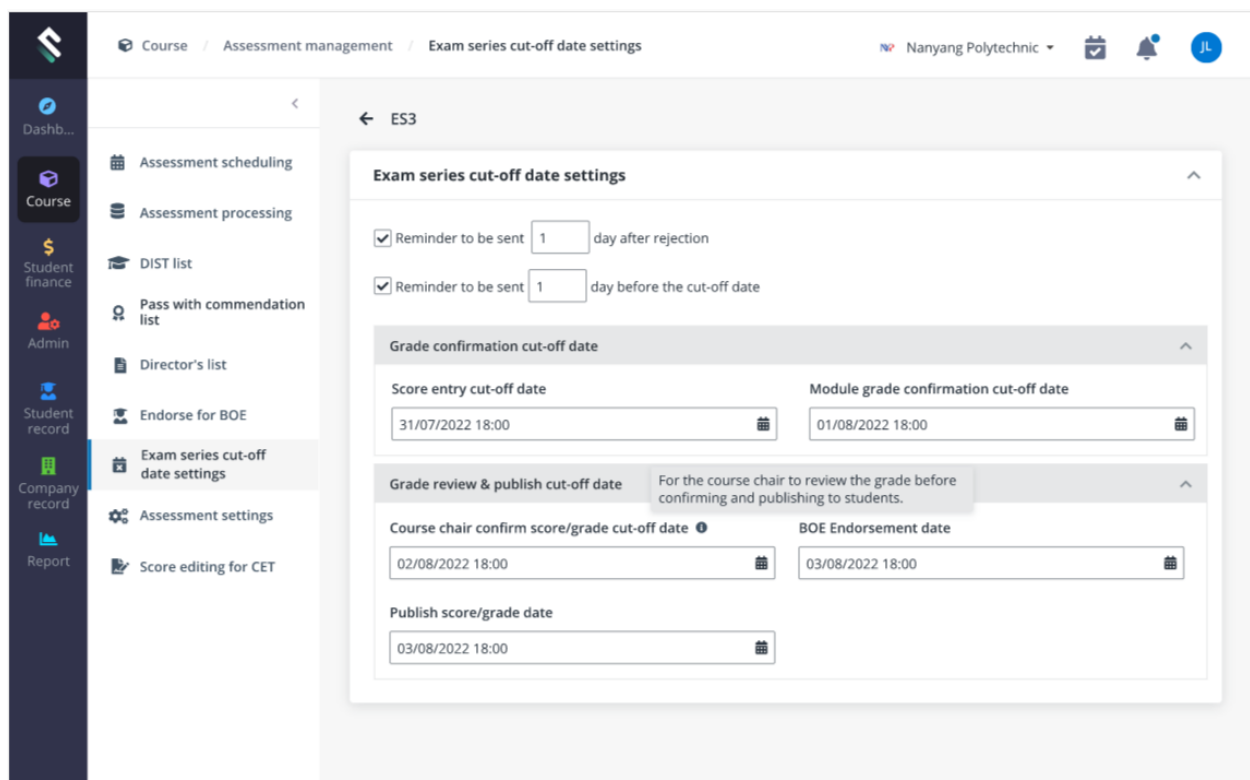
Assessment status: All

Search by student name...

Export Grade moderation **Reject**

☐	Student name	Student ID	Course intake No.	PDC/MC code	Class	Mod
☐	VC Valentin Connolly	A001	NYP-0427-2 20046	Core 1	Class 1	97
☐	DL Declan Lehner	A002	NYP-0427-2 20046	Core 1	Class 1	97
☐	EG Eulah Glover	A003	NYP-0427-2 20046	Core 1	Class 1	93

- For any changes to the grades/score for assessment, the user would have to complete the approval process flow in sTMS and follow the cut-off date settings on sTMS.



Course / Assessment management / Exam series cut-off date settings

Nanyang Polytechnic

ES3

Exam series cut-off date settings

☒ Reminder to be sent 1 day after rejection

☒ Reminder to be sent 1 day before the cut-off date

Grade confirmation cut-off date

Score entry cut-off date: 31/07/2022 18:00

Module grade confirmation cut-off date: 01/08/2022 18:00

Grade review & publish cut-off date

For the course chair to review the grade before confirming and publishing to students.

Course chair confirm score/grade cut-off date: 02/08/2022 18:00

BOE Endorsement date: 03/08/2022 18:00

Publish score/grade date: 03/08/2022 18:00

2.4.3 Email notification for each upload

An email notification will send to <ITE Admin email address> to report the records process details after sTMS successfully handled OMR http request (status code=200). Email template as follows:

Subject: YYYYMMDDhhmm DOSS – sTMS upload details
Attachment: YYYYMMDDhhmm DOSS Upload to sTMS.csv
Body:
Dear Officer,
Results have been uploaded to sTMS from ITE DOSS system.
Statistics- {success count} successful, {error count} error of total {total count} records.

Possible error messages for record level are listed as follows:

SN	Error message
1	Invalid format for '{field name}'.
2	No '{field name}' value found.
3	Results cannot be uploaded because current status of results is "confirmed". <i>(Existence of confirmed results which cannot be overwritten unless status of results is changed from 'confirmed' to 'unsubmitted')</i>
4	Upload date later than score-entry cut-off date.

Sample attachment csv file:



202209271627 DOSS
upload to sTMS.csv

3. Interface Fields

3.1 Interface launch parameter

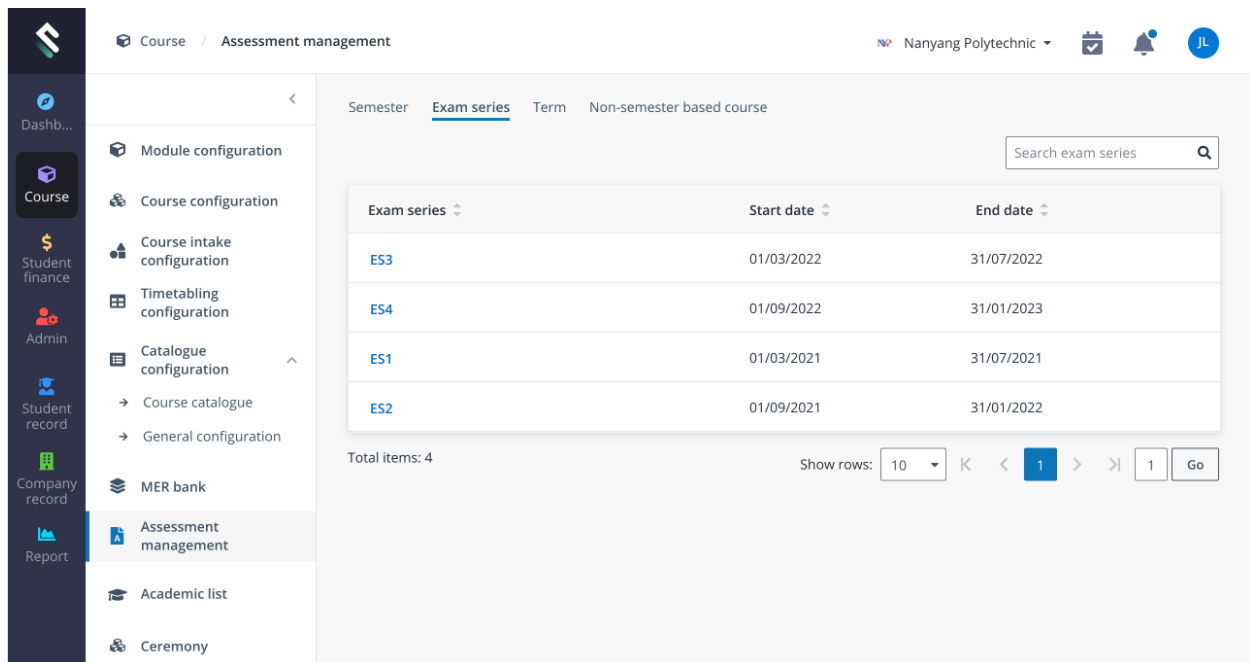
sTMS Field Name	OMR Scanning Field Name	Description	Data Type	Length	Remarks
StudentId*	StudentId	UAM Common ID	String	20	
ExamSeries	ExamSeries	Exam Series	String	20	Sample: YYYYMM
Subcomponent*	Assignment Item	Subcomponent name	String	20	Sample: CLT1, TEXM, PEXM
ModuleCode*	ModuleCode	Module code	String	255	Sample: BM0001 <i>*Note that this module code shall be unique for identification purposes for semester/non-semester courses.</i>
Score*	Mark	Assessment score	Number	-	Sample: 30 If the value is 999, it means the student is absent
ModifiedBy*	StaffId	Staff ID using AAD Email Login	String	50	AAD Email login for upload by users, rescoring would be indicated as DOSSRESCR2. For marks not recorded from scanning, Staff ID shall be indicated as ETUTOR2 from OMR Scanning System.
ModifiedTime*	SFTPDateTime	Date of creation or modified date	Datetime	50	Sample: yyyy-MM-dd hh:mm:ss
CriticalSkills	Cpass	Critical Skills Note: This field is not a compulsory setting.	String	255	Sample: P, F, '-' or NA

*represents compulsory fields to be sent.

3.2 Interface Response Data Fields

sTMS Field Name	Description	Data Type	Length	Remarks
StudentId	UAM Common ID	String	20	
ExamSeries	Exam Series	String	20	Format: YYYYMM
Subcomponent	Subcomponent name	String	20	
ModuleCode	Module code	String	255	
IntakeNo	Course intake number	String	20	
ClassName	Class name	String	20	
Successful	Upload status of current record item.	Boolean	-	True – Success False – Failure
ErrorMessage	Error message	String	255	Error message to explain why current record not uploaded successfully. Empty if upload successfully.
Timestamp	Timestamp marked by sTMS	Datetime	-	
Modifiedby	Staff ID using AAD Email Login	String	50	AAD Email login for initial upload by users, rescoring would be indicated as DOSSRESCR2.

Please note that exam series would be indicated as YYYYMM.



The screenshot shows the 'Assessment management' interface for Nanyang Polytechnic. The left sidebar contains navigation links: Dashboard, Course, Student finance, Admin, Student record, Company record, and Report. The main content area is titled 'Assessment management' and includes a search bar for 'Exam series'. Below the search bar, there is a table with columns: Exam series, Start date, and End date. The table displays four rows of data:

Exam series	Start date	End date
ES3	01/03/2022	31/07/2022
ES4	01/09/2022	31/01/2023
ES1	01/03/2021	31/07/2021
ES2	01/09/2021	31/01/2022

Below the table, it indicates 'Total items: 4' and provides pagination controls: 'Show rows: 10', '1', and 'Go'.

 Course / Assessment management / Assessment processing / Module details

Nanyang Polytechnic

Dashb...

Course

Student finance

Admin

Student record

Company record

Report

Assessment scheduling

Assessment processing

DIST list

Pass with commendation list

Director's list

Endorse for BOE

Semester cut off date settings

Assessment settings

Score editing for CET

BM0001- Business management

Module

Module overview

Course work

Component overview

CW 1

CW 2

Final exam

Final exam

← BM0001 - e-Business Management - CW1

Base score
100

Weightage(%)
20%

Grade mapping rule
Grade mapping rule 1

Unsubmitted
Pending approval
Confirmed
All

Assessment status: All

Search by student name...

Edit
Import
Export

le	Assessment status	Modified on	Modified by	Remarks
	XM			
	DC	3/5/2022 18:05	Eric Zheng	
		3/5/2022 18:05	Eric Zheng	

Total items: 4

Show rows: 10

1

Go

 Course / Assessment management / Assessment processing / Module details / Edit

← CW1

Assessment status: All

Search by student name...

Student name	Student ID	Class	Score	Given score	Mapped grade	Given grade	Pass critical skill	Assessment status	Remarks
VC Valentin Connolly	A001	Class 1						XM	
DL Declan Lehner	A002	Class 1	90.9	90.9	A	A		DC	
EG Eulah Glover	A003	Class 2	69.1	69.1	D	D			
DF Devin Fisher	A004	Class 2							

Total items: 4

Show rows: 10

1

Go

Close

Save

3.3 Sample Responses

Successful Operation Sample (Status Code 200)

```
{
  "data": [
    {
      "StudentId": "S1234567",
      "ExamSeries": "202205",
      "Subcomponent": "CLT",
      "ModuleCode": "BM0001",
      "IntakeNo": "BM0001-1",
      "ClassName": "Class A",
      "Successful": "True",
      "ErrorMessage": "",
      "Timestamp": "2022-05-24 16:00:00",
      "Modifiedby": "EricZheng@AADEmail.com.sg"
    },
    {
      "StudentId": "S7654321",
      "ExamSeries": "@202205",
      "Subcomponent": "CLT",
      "ModuleCode": "BM0001",
      "IntakeNo": "BM0001-1",
      "ClassName": "Class A",
      "Successful": "False",
      "ErrorMessage": "Invalid field format for 'ExamSeries'.",
      "Timestamp": "2022-05-24 16:00:00",
      "Modifiedby": "DOSSRESCR2"
    }
  ]
}
```

```
    }  
  ]  
}  
  
Bad Request Sample (Status code 400)  
  
{  
  "Request body is invalid"  
}  
  
Internal Server Error (Status code 500)  
  
{  
  "Internal server error."  
}
```


4. Status Codes

The table below provides the detailed error listing and handling procedure. These status codes would be response to client in the Http response header.

Status Code	Message	Description	Handling Procedure
200	NA	Successful Request <i>(Please note that possible error messages would be included in csv file for each upload. Refer to Section 2.4.3 Email notification for each upload for more details)</i>	NA
400	Bad request	Request body is invalid <i>(This is a generic error code which indicates request body is not recognized, not to indicate a record level exception.)</i>	Verify the interface launch parameter in your HTTP request; Raise a ticket to sTMS support team for further investigation.
500	Internal server error	Maintenance, system down	sTMS support team will inform ITE system admin if schedule maintenance. If error occur during working hour, Raise a ticket to sTMS support team for further investigation.

5. Interface Schemas

Interface Schemas are detailly described in the following table:

OMRResult ▾ {		←
StudentId	string	UAM Common ID
ExamSeries	string	Exam Series
Subcomponent	string	Subcomponent name
ModuleCode	string	Module code
Score	number	Assessment score
ModifiedBy	string	Staff ID
ModifiedTime	string(\$yyyy-MM-dd hh:mm:ss)	
CriticalSkills	string	Critical Skills
	Enum:	
	☐	[P, F]
}		